TYLER GUEST

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EDUCATION

University of Michigan-Flint Dec 2027 (Expected)

- B.S. Computer Science (GPA: 3.86/4.0)
- Relevant Coursework: Object-Oriented Programming, Data Structures, Parallel & Distributed Processing, Computer Architecture, Operating Systems, Theory of Computation, Discrete Structures, Database Design

TECHNICAL PROJECTS

Infurnace | [github] Aug 2026 - Present

Python, pytest, tinygrad

- Built a local LLM inference application using tinygrad, supporting validated GGUF loading, automatic device selection, context-window management, and streaming generation.
- Adapted and integrated GPU kernels for packed Q4_K/Q6_K matrix-vector operations with Q8 activation quantization, achieving ~93 tokens/s in short Qwen3-4B decode benchmarks on an RTX 5060.
- Developed automated tests covering model loading, tokenization, generation limits, cache reuse, and numerical parity between eager and JIT kernel execution.

Memoman | [github] Sep 2025 - Present

C, Make

- Built a TLSF-inspired memory allocator in C with O(1) free-block lookup using two-level bitmaps and segregated free lists, plus adjacent-block coalescing.
- Implemented a comprehensive test suite and randomized stress test covering 500,000+ allocation, free, reallocation, and aligned allocation operations with zero failures and clean AddressSanitizer/UBSan results.
- Benchmarked against reference TLSF and glibc malloc on reproducible single-threaded workloads, reaching ~21M ops/s on random churn and ~7% higher median throughput than glibc across five 30-second runs.

Service & Inventory Database | [github] Sep 2025 - Dec 2025

SQL

- Designed a normalized relational database connecting customers, service tickets, estimates, invoices, products, inventory, payments, and repair-status records.
- Implemented PL/SQL triggers to generate invoice line items, update inventory, and automatically recalculate invoice subtotal, tax, and total.
- Created reporting queries for invoice details, unpaid balances, parts usage, ticket-status counts, and customer spending.

WORK EXPERIENCE

We Can Do It Graphics Flint, MI

Lead Graphic Designer May 2015 - Jan 2023

- Led a five-person team in a high-volume production environment, coordinating workloads, quality control, and production priorities.
- Operated and troubleshot industrial printing equipment while maintaining production logs, inventory records, and job-tracking documentation.
- Reviewed vector artwork, dimensions, materials, and production specifications before manufacturing, preventing errors and ensuring compliance with customer requirements.

SKILLS

- **Languages:** C, C++, Python, SQL
- **Systems & Tools:** Linux, Git, Make
- **Libraries & Frameworks:** tinygrad, pytest